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Journal of the Franklin Institute 358 (2021) 4793–4806

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High-gain fractional disturbance observer control of uncertain dynamical systems

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Received 11 December 2019; received in revised form 9 December 2020; accepted 6 April 2021
Available online 15 April 2021

Abstract

Disturbance observer-based control allows to compensate unknown inputs, however, in most cases, requiring their integer-order differentiability. In this paper, a novel disturbance observer-based state feedback controller is proposed to compensate a more general class of fractional-, but not necessarily integer-order, differentiable unknown inputs. The proposed fractional PI-like structure yields precise conditions for feedback gain tuning. Remarkably, the resulting controller rejects non-differentiable disturbances with a smooth controller, guaranteeing robustness, an outstanding features for tracking tasks, under a prescribed practical stability regimen. A comparison to a fractional sliding mode observer is conducted via simulations to highlight the reliability of the proposed scheme.
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1. Introduction

The high-gain concept was pioneered in [1–3] addressing observer design, assuring precise and fast state estimation, later extended into high-gain observer-based control in [4–9]. These

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https://doi.org/10.1016/j.jfranklin.2021.04.020
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schemes arose as viable options for robust and fast estimation throughout increasing the feedback gains, as long as the controller remains within admissible bounds. Although high-gain controllers induce only practical stability [10], they facilitate implementation and tuning effort, in contrast to sound robust controllers, such as variable structure and model-based schemes, in which the theoretical asymptotic and finite-time stability are not achievable in practice.

Disturbance observer-based control is considered in [11,12] based on a system nominal model without uncertainties nor disturbances. Frequency domain-based designs for linear systems have shown advantages in practice since relevant performance can be set, including bandwidth constraints. For nonlinear disturbance observers, lumped disturbances are considered as a typical approach [13,14]. Recently, high-gain disturbance observers have been proposed for integer-order systems [15–18], guaranteeing prescribed tracking. All these results assume that the unknown lumped disturbance is Lipschitz continuous, which for some advanced applications, could be a limitation. In response to this issue, fractional sliding mode control has been considered with a disturbance observer [19]. However, there remains a question on how to deal with simpler control structures with fractional disturbance observers. In this direction, Martínez-Fuentes and Martínez-Guerra [20] proposed a Mittag–Leffler stable state observer for fractional-order systems.

The contribution of this paper is then, the formulation of [20] in the realm of [21,22] to provide a smooth controller able to precisely compensate continuous but not necessarily differentiable disturbances.

Fractional calculus constitutes an important and emerging tool to address advanced and complex phenomena [23], including a large variety of engineering systems subject to a wider class of dynamic effects [24]. Fractional-order controllers have been widely considered since they significantly enhance the dynamical performance of their integer-order counterparts [25,26]. In [21,22], sliding mode-based disturbance observers are studied to account for lumped Hölder continuous disturbances, that is, not necessarily integer-order differentiable functions, which possess well-posed fractional-order derivatives. Later Muñoz-Vázquez et al. [27] extended this idea to provide a theoretically finite-time exact estimation of Hölder continuous disturbances, nonetheless, the unavoidable chattering phenomenon affects the tracking precision, as well as the integrity of the system components in real-time applications. Other approaches have combined disturbance observers and fractional-order tools [28–34].

In contrast to the outstanding and inspiring mentioned contributions, the approach proposed in this paper stands for: *A high-gain smooth PI-like structure to observe continuous but not necessarily differentiable disturbances, whose tracking precision can be adjusted with a single control parameter. In this sense, the proposed scheme provides a methodology that modulates constant gains through adjusting the ultimate upper bound of the tracking errors, thus providing further insights on the tuning procedure.*

The rest of this paper is organized as follows. Section 2 presents basic preliminaries on fractional-order systems and stability analysis. Section 3 introduces the system definition and the high-gain observer based control design. Section 4 presents a simulation study. Finally, conclusive discussions are presented in Section 5.

2. Preliminaries

2.1. On fractional-order systems

Fractional calculus deals with differentiation and integration of non-integer orders by means of differintegral operators. For $\boldsymbol{x} : \Omega_t \to \mathbb{R}^n$ a real-valued vector function and $\Omega_t \subset$

$\mathbb{R}$ a compact set, the differintegral operators of order $\alpha \in (0,1)$ are defined as follows [23,35,36]:

- *Riemann–Liouville integral*

$$\mathscr{I}^{\alpha}\boldsymbol{x}(t) = \frac{1}{\Gamma(\alpha)}\int_0^t \frac{\boldsymbol{x}(\tau)}{(t-\tau)^{1-\alpha}}d\tau \tag{1}$$

- *Marchaud derivative*

$$^{M}\mathscr{D}^{\alpha}\boldsymbol{x}(t) = \frac{\boldsymbol{x}(t)}{\Gamma(1-\alpha)(t-t_0)^{\alpha}} + \frac{\alpha}{\Gamma(1-\alpha)}\int_0^t \frac{\boldsymbol{x}(t)-\boldsymbol{x}(\tau)}{(t-\tau)^{\alpha+1}}\,d\tau \tag{2}$$

- *Extended Caputo derivative*

$$\mathscr{D}^{\alpha}\boldsymbol{x}(t) = \frac{\boldsymbol{x}(t)-\boldsymbol{x}(0)}{\Gamma(1-\alpha)(t-t_0)^{\alpha}} + \frac{\alpha}{\Gamma(1-\alpha)}\int_0^t \frac{\boldsymbol{x}(t)-\boldsymbol{x}(\tau)}{(t-\tau)^{\alpha+1}}\,d\tau \tag{3}$$

where $\Gamma(\varsigma) = \int_0^{\infty} z^{\varsigma-1}e^{-z}dz$ is the Gamma function; accordingly, it follows that

$$\mathscr{D}^{\alpha}\boldsymbol{x}(t) = {}^{M}\mathscr{D}^{\alpha}[\boldsymbol{x}(t)-\boldsymbol{x}(0)]. \tag{4}$$

Furthermore, in the case of $\boldsymbol{x}(t)$ is weakly integer-order differentiable, one has that Eqs. (2) and (3) become

$$\begin{aligned} {}^{M}\mathscr{D}^{\alpha}\boldsymbol{x}(t) &= {}^{RL}\mathscr{D}^{\alpha}\boldsymbol{x}(t) = \frac{d}{dt}\mathscr{I}^{1-\alpha}\boldsymbol{x}(t), \\ \mathscr{D}^{\alpha}\boldsymbol{x}(t) &= {}^{C}\mathscr{D}^{\alpha}\boldsymbol{x}(t) = \mathscr{I}^{1-\alpha}\dot{\boldsymbol{x}}(t), \end{aligned} \tag{5}$$

where ${}^{RL}\mathscr{D}^{\alpha}\boldsymbol{x}(t)$ and ${}^{C}\mathscr{D}^{\alpha}\boldsymbol{x}(t)$ are the classical Riemann–Liouville and Caputo derivatives, and integration by parts of Eqs. (2) and (3) is used. It is worth noticing that the Marchaud operator does not require the use of an integer-order derivative in its definition. However, Marchaud and Riemann–Liouville derivatives coincide for sufficiently good functions. Besides, it has been proved that [27,35],

$$\begin{aligned} {}^{M}\mathscr{D}^{\alpha}\mathscr{I}^{\alpha}\boldsymbol{x}(t) &= \boldsymbol{x}(t), \\ \mathscr{I}^{\alpha}\mathscr{D}^{\alpha}\boldsymbol{x}(t) &= \boldsymbol{x}(t)-\boldsymbol{x}(0). \end{aligned} \tag{6}$$

for sufficiently good, but not necessarily integer-order differentiable, real-valued vector function $\boldsymbol{x}(t)$.

For two real-valued and Lebesgue integrable functions $x(t)$ and $y(t)$, with $x(t) \le y(t)$ for almost all $t \ge 0$, one has that

$$\mathscr{I}^{\alpha}x(t) \le \mathscr{I}^{\alpha}y(t)\ \forall t \ge 0, \text{ whenever } \alpha > 0. \tag{7}$$

The following definition from [37] is also of preponderant interest.

Definition 1. The constant vector $\boldsymbol{x}^* \in \mathbb{R}^n$ is an equilibrium point for $\mathscr{D}^{\alpha}\boldsymbol{x} = \boldsymbol{f}(t,\boldsymbol{x})$ if $\mathscr{D}^{\alpha}\boldsymbol{x}^* = \boldsymbol{f}(t,\boldsymbol{x}^*)$ for all $t \ge 0$.

Since the extended Caputo derivative of a constant is null, $\boldsymbol{x}^*$ is an equilibrium if and only if $\boldsymbol{f}(t,\boldsymbol{x}^*) = \boldsymbol{0}$ for all $t \ge 0$. Therefore, $\boldsymbol{y}(t) = \boldsymbol{0}$ is an equilibrium for $\mathscr{D}^{\alpha}\boldsymbol{y}(t) = \bar{\boldsymbol{f}}(t,\boldsymbol{y}(t))$, with $\boldsymbol{y}(t) = \boldsymbol{x}(t) - \boldsymbol{x}^*$ and $\bar{\boldsymbol{f}}(t,\boldsymbol{y}(t)) = \boldsymbol{f}(t,\boldsymbol{y}(t)+\boldsymbol{x}^*)$.

The Mittag–Leffler stability generalizes the exponential stability of the integer-order case to the more general fractional-order case. Thus, consider the following definition [37].

Definition 2. The solution of $\mathscr{D}^{\alpha}\boldsymbol{x}(t) = f(t, \boldsymbol{x}(t))$ is Mittag–Leffler stable if

$$\|\boldsymbol{x}(t)\| \leq \left\{m(\boldsymbol{x}(0))E_{\beta}(-at^{\beta})\right\}^{b}$$

for all $t \geq 0$, where $m(\boldsymbol{x}) \geq 0$ is a locally Lipschitz continuous function on $\boldsymbol{x}$ with $m(\boldsymbol{0}) = 0$, for $\beta \in (0, 1)$ and some real numbers $a, b > 0$, where

$$E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + \beta)} \tag{8}$$

is the two-parameters Mittag–Leffler function, with $\alpha, \beta > 0$. Besides, one has that $e^z = E_{1,1}(z)$ is the exponential function, and $E_{\alpha,1}(z) = E_{\alpha}(z)$ is the one-parameter Mittag–Leffler function.

The stability of fractional-order nonlinear systems has been studied in [8], and the Mittag–Leffler stability of a fractional-order differintegral inequality was proved in [38,39] for quadratic functions. More recently, Muñoz-Vázquez et al. [36] demonstrated that the same inequality of [38,39] is valid for operator (3), and the validity of such inequality in the case of nonsmooth convex functions was demonstrated [40]. Thus, consider the following.

Definition 3 (Definition 3.1.1 of [41], p. 112)**.** A function $V : \Omega_x \subset \mathbb{R}^n \to \mathbb{R}$ is convex if Ω_x is convex and $\forall \boldsymbol{x}, \boldsymbol{y} \in \Omega_x$, the following inequality holds for all $\lambda \in [0, 1]$,

$$V(\lambda\boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leq \lambda V(\boldsymbol{x}) + (1-\lambda)V(\boldsymbol{y}). \tag{9}$$

Definition 4 (Corollary 2.6 of [42], p. 35)**.** Let $V : \Omega_x \subset \mathbb{R}^n \to \mathbb{R}$ be a convex function. The set-valued map

$$\partial V(\boldsymbol{x}) = \left\{\boldsymbol{\zeta} \in \mathbb{R}^n : \ V(\boldsymbol{y}) - V(\boldsymbol{x}) - \boldsymbol{\zeta}^T(\boldsymbol{y} - \boldsymbol{x}) \geq 0, \quad \forall \boldsymbol{y} \in \Omega_x\right\} \tag{10}$$

is the proximal subdifferential of V at $\boldsymbol{x} \in \Omega_x$, and $\boldsymbol{\zeta}(\boldsymbol{x}) \in \partial V(\boldsymbol{x})$ is a proximal subgradient of V at $\boldsymbol{x}$.

In the case of V is differentiable at $\boldsymbol{x}$, one has that $\partial V(\boldsymbol{x}) = \{\nabla V(\boldsymbol{x})\}$, and thus, Eq. (10) becomes

$$\nabla V(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) \leq V(\boldsymbol{y}) - V(\boldsymbol{x}) \tag{11}$$

The following Lemma and Theorem of [40] are of interest for the control methods proposed in this paper.

Lemma 1. *Let $V(\boldsymbol{x}(t))$ be a real-valued function that is Lipschitz continuous on $\boldsymbol{x}$, with convex domain $\Omega_x \subseteq \mathbb{R}^n$, and let $\boldsymbol{x} \in \Omega_x$ be a real-valued vector function whose entries are continuous functions with well-posed fractional derivatives of at least some order $\alpha \in (0, 1)$. If $V(\boldsymbol{x}(t))$ is convex, then*

$$\mathscr{D}^{\alpha}V(\boldsymbol{x}(t)) \leq \inf_{\boldsymbol{\zeta}(\boldsymbol{x}) \in \partial V(\boldsymbol{x})} \boldsymbol{\zeta}(\boldsymbol{x}(t))^T \mathscr{D}^{\alpha}\boldsymbol{x}(t). \tag{12}$$

Theorem 1. *Let $V(\boldsymbol{x}(t))$ be a positive definite real-valued function, which is Lipschitz continuous and convex in $\boldsymbol{x}$, and $\boldsymbol{0} \in \partial V(\boldsymbol{0})$. If*

$$\inf_{\boldsymbol{\zeta}(\boldsymbol{x}) \in \partial V(\boldsymbol{x})} \boldsymbol{\zeta}(\boldsymbol{x}(t))^T \mathscr{D}^{\alpha}\boldsymbol{x}(t) \leq -\gamma V(\boldsymbol{x}(t))^q \tag{13}$$

for almost all $t \geq 0$, $\gamma > 0$, and $q \in (0, 1]$, then $V(\boldsymbol{x}(t))$ is Mittag–Leffler stable. If in addition $V(\boldsymbol{x}(t)) = \|\boldsymbol{x}(t)\|^b$ for some $b > 0$, then $\|\boldsymbol{x}(t)\|$ is also Mittag–Leffler stable.

3. Controller design

3.1. System definition

Consider the following linear time invariant (LTI) system subject to a wide class of not necessarily integer-order differentiable disturbances, non-linearities and dynamic uncertainties,

$$\dot{\boldsymbol{x}}(t) = A\boldsymbol{x}(t) + B\boldsymbol{u}(t) + E\boldsymbol{d}(t) \tag{14}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ is the state, which is assumed available for control purposes, $\boldsymbol{u} \in \mathbb{R}^{n_u}$ is the control input, $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times n_u}$ are constant matrices, $\boldsymbol{d} \in \mathbb{R}^{n_d}$ is an unknown but continuous disturbance, with $E \in \mathbb{R}^{n\times n_d}$ a constant matrix, and, without loss of generality, consider that $n_u, n_d \leq n$. The following standard assumptions on system (14) are considered [43],

- **Assumption 1.** The pair (A, B) is controllable.
- **Assumption 2.** The matrix B has full-column rank.
- **Assumption 3.** The disturbance is matched, that is, $E = B$.
- **Assumption 4.** The disturbance $\boldsymbol{d}$ satisfies $\sup \|\mathscr{D}^{\alpha}\boldsymbol{d}\| < \infty$ for $\alpha \in (0, \alpha_c)$, and $\alpha_c > 0$ the critical order of $\boldsymbol{d}$.

The family of disturbances is fully characterized by Assumption 4, this means that impulsive or very high frequency disturbances are not considered. The scalar components of those real-valued vector functions that satisfy Assumption 4 can be characterized by the Hölder condition [44,45], that is, $f(t)$ is α_c-Hölder continuous on Ω if there exists a constant H such that $|f(t_2) - f(t_1)| \leq H|t_2 - t_2|^{\alpha_c}$, for some $\alpha_c \in (0, 1)$. In this sense, $f(t)$ posses uniformly continuous derivatives of any order $\alpha \in (0, \alpha_c)$. It is worth to comment that Hölder continuity is less restrictive than Lipschitz condition, which only considers the case of $\alpha_c = 1$. In addition, Lipschitz continuity is commonly assumed during control design [7].

3.2. Disturbance observer-based control design

In accordance to Assumption 1, there is a matrix $K \in \mathbb{R}^{n\times n_u}$, such that the eigenvalues of $\bar{A} = A - BK$ are located in the open left-half plane. Therefore, the ideal controller

$$\boldsymbol{u}_0(t) = -K\boldsymbol{x}(t) \tag{15}$$

produces $\boldsymbol{x} \to \boldsymbol{0}$ when $t \to \infty$, in the case of $\boldsymbol{d} = \boldsymbol{0}$ and $\boldsymbol{u} = \boldsymbol{u}_0$. Thus, the controller is proposed as $\boldsymbol{u}(t) = \boldsymbol{u}_0(t) - \hat{\boldsymbol{d}}(t)$, this is,

$$\boldsymbol{u}(t) = -K\boldsymbol{x}(t) - \hat{\boldsymbol{d}}(t) \tag{16}$$

for $\hat{\boldsymbol{d}}(t)$ an observer of disturbance $\boldsymbol{d}(t)$. Moreover, by considering the controller (16), Assumption 3 implies that system (14) can be rewritten as

$$\dot{\boldsymbol{x}}(t) = \bar{A}\boldsymbol{x}(t) + B\Big(-\hat{\boldsymbol{d}}(t) + \boldsymbol{d}(t)\Big). \tag{17}$$

Consider the integral auxiliary variable

$$\boldsymbol{\sigma}_0(t) = B^{+}\left(\boldsymbol{x}(t) - \boldsymbol{x}(0) - \int_0^t \bar{A}\boldsymbol{x}(\tau)d\tau\right) \tag{18}$$

whose first-order derivative complies with

$$\dot{\boldsymbol{\sigma}}_0(t) = -\hat{\boldsymbol{d}}(t) + \boldsymbol{d}(t). \tag{19}$$

In addition, let $\boldsymbol{\sigma}$ be a differintegral variable defined as

$$\boldsymbol{\sigma}(t) = \mathscr{D}^{1-\alpha}\boldsymbol{\sigma}_0(t). \tag{20}$$

In virtue of $\boldsymbol{\sigma}_0(0) = \mathbf{0}$, the operator $\mathscr{D}^{1-\alpha}$ can be either the extended Caputo or the Marchaud derivative. Furthermore, the differentiability of $\boldsymbol{\sigma}_0$ implies that

$$\begin{aligned}\mathscr{D}^{\alpha}\boldsymbol{\sigma}(t) &= \dot{\boldsymbol{\sigma}}_0(t),\\ &= -\hat{\boldsymbol{d}}(t) + \boldsymbol{d}(t).\end{aligned} \tag{21}$$

Thus, taking into account the fractional-order differentiability of the disturbance, the observer is proposed as the high-gain fractional PI

$$\hat{\boldsymbol{d}}(t) = \frac{k_p}{\varepsilon}\boldsymbol{\sigma}(t) + \frac{k_i}{\varepsilon^2}\mathscr{I}^{\alpha}\boldsymbol{\sigma}(t). \tag{22}$$

for k_p and k_i positive scalar feedback gains, which satisfy that, the roots of the polynomial $\rho(s) = s^2 + k_p s + k_i = 0$ are located in the open left-half plane, this is $k_p, k_i > 0$. The parameter ε is designed to adjust the precision of $\hat{\boldsymbol{d}} \to \boldsymbol{d}$.

The change of variables

$$\begin{aligned}\boldsymbol{\xi}_1 &= \frac{\boldsymbol{\sigma}}{\varepsilon}\\ \boldsymbol{\xi}_2 &= -\frac{k_i}{\varepsilon^2}\mathscr{I}^{\alpha}\boldsymbol{\sigma} + \boldsymbol{d}\end{aligned} \tag{23}$$

produces

$$\begin{aligned}\varepsilon\mathscr{D}^{\alpha}\boldsymbol{\xi}_1(t) &= -k_p\boldsymbol{\xi}_1(t) + \boldsymbol{\xi}_2(t).\\ \varepsilon\mathscr{D}^{\alpha}\boldsymbol{\xi}_2(t) &= -k_i\boldsymbol{\xi}_1(t) + \varepsilon\mathscr{D}^{\alpha}\boldsymbol{d}(t),\end{aligned} \tag{24}$$

or in a condensed form

$$\varepsilon\mathscr{D}^{\alpha}\boldsymbol{\xi}(t) = \Lambda\boldsymbol{\xi}(t) + \varepsilon\boldsymbol{\delta}(t) \tag{25}$$

with $\boldsymbol{\xi}^T = [\boldsymbol{\xi}_1^T \quad \boldsymbol{\xi}_2^T]$, $\Lambda = \begin{bmatrix} -k_p I & I \\ -k_i I & 0 \end{bmatrix}$, and $\boldsymbol{\delta}^T = [\mathbf{0}^T \quad \mathscr{D}^{\alpha}\boldsymbol{d}^T]$.

3.3. Stability analysis

Since the scalar components of the differential equations that conform system (24) are completely decoupled, consider without loss of generality that $n_u = 1$, that is, $\boldsymbol{\xi} = [\xi_1 \quad \xi_2]^T$, $\Lambda = \begin{bmatrix} -k_p & 1 \\ -k_i & 0 \end{bmatrix}$, and $\boldsymbol{\delta} = [0 \quad \mathscr{D}^{\alpha}d]^T$. Thus, the stability properties of system (25) are analyzed in the following main result.

Theorem 2. *Consider the fractional-order system (25). Then, for k_p, k_i and $\varepsilon > 0$, the disturbance estimation error $\Delta d = d - \hat{d}$ converges inside an open ball centered at the origin, whose radius is proportional to ε.*

Proof. Since Λ is a Hurwitz matrix, there exists symmetric and positive definite matrices $P, Q \in \mathbb{R}^{2\times 2}$ such that

$$2Q + P\Lambda + \Lambda^T P = 0. \tag{26}$$

Now, consider the candidate Lyapunov function

$$V = \varepsilon \|P_0 \boldsymbol{\xi}\| \tag{27}$$

where P_0 satisfies $P = P_0^T P_0$, and $||\cdot||$ is the conventional Euclidean vector norm. Then, the fractional derivative of V complies to

$$\begin{aligned}
\mathscr{D}^\alpha V &\leq \varepsilon \frac{\boldsymbol{\xi}^T P}{\|P_0\boldsymbol{\xi}\|} \mathscr{D}^\alpha \boldsymbol{\xi} \\
&= \frac{1}{\|P_0\boldsymbol{\xi}\|}\left(-\boldsymbol{\xi}^T Q\boldsymbol{\xi} + \varepsilon \boldsymbol{\xi}^T P\boldsymbol{\delta}\right) \\
&\leq \frac{1}{\|P_0\boldsymbol{\xi}\|}\left(-\lambda_m(Q)\|\boldsymbol{\xi}\|^2 + \varepsilon\|P_0\boldsymbol{\xi}\|\lambda_M(P_0)\|\boldsymbol{\delta}\|\right) \\
&= -\frac{\lambda_m(Q)\|\boldsymbol{\xi}\|^2}{\|P_0\boldsymbol{\xi}\|} + \varepsilon\lambda_M^{1/2}(P)\|\boldsymbol{\delta}\|
\end{aligned} \tag{28}$$

for $\lambda_m(\cdot)$ and $\lambda_M(\cdot)$ the minimum and maximum singular value operators.

Now, considering that $\|P_0\boldsymbol{\xi}\|^2 = \boldsymbol{\xi}^T P\boldsymbol{\xi}$, one has that $\lambda_m(P)\|\boldsymbol{\xi}\|^2 \leq \|P_0\boldsymbol{\xi}\|^2 \leq \lambda_M(P)\|\boldsymbol{\xi}\|^2$, consequently one obtains

$$\mathscr{D}^\alpha V \leq -\gamma V + \varepsilon\lambda_M^{1/2}(P)k_\delta. \tag{29}$$

where $\gamma = \dfrac{\lambda_m(Q)}{\varepsilon\lambda_M(P)}$ and $k_\delta > \|\boldsymbol{\delta}\|$. Then, integrating Eq. (29) leads to

$$V(t) \leq V(0)E_\alpha(-\gamma t^\alpha) + \varepsilon\lambda_M^{1/2}(P)k_\delta \int_0^t \tau^{\alpha-1}E_{\alpha,\alpha}(-\gamma\tau^\alpha)d\tau. \tag{30}$$

Evaluating the integral in the above inequality leads to

$$\begin{aligned}
\int_0^t \tau^{\alpha-1}E_{\alpha,\alpha}(-\gamma\tau^\alpha)d\tau &= \int_0^t \sum_{k=0}^{\infty} \frac{(-\gamma)^k \tau^{(k+1)\alpha-1}}{\Gamma((k+1)\alpha)} d\tau \\
&= \sum_{k=0}^{\infty} \frac{(-\gamma)^k t^{(k+1)\alpha}}{\Gamma((k+1)\alpha+1)} \\
&= (-\gamma)^{-1} \sum_{k=1}^{\infty} \frac{(-\gamma)^k t^{k\alpha}}{\Gamma(k\alpha+1)} \\
&= (-\gamma)^{-1}\left[\sum_{k=0}^{\infty} \frac{(-\gamma)^k t^{k\alpha}}{\Gamma(k\alpha+1)} - 1\right] \\
&= \gamma^{-1}\left[1 - E_\alpha(-\gamma t^\alpha)\right] \leq \gamma^{-1}
\end{aligned} \tag{31}$$

where the term $E_\alpha(-\gamma t^\alpha)$ vanishes and is completely monotonic, Theorem 4.4 of [46], p. 71. Therefore

$$V(t) \leq V(0)E_\alpha(-\gamma t^\alpha) + \frac{\varepsilon\lambda_M^{1/2}(P)k_\delta}{\gamma}$$

$$= V(0)E_\alpha(-\gamma t^\alpha) + \frac{\varepsilon^2 k_\delta \lambda_M^{3/2}(P)}{\lambda_m(Q)}. \tag{32}$$

Now, using

$$\varepsilon\lambda_m^{1/2}(P)\|\boldsymbol{\xi}(t)\| \le V(t) \le \varepsilon\lambda_M^{1/2}(P)\|\boldsymbol{\xi}(t)\|, \tag{33}$$

one has that

$$\|\boldsymbol{\xi}(t)\| \le \frac{\lambda_M^{1/2}(P)}{\lambda_m^{1/2}(P)}\|\boldsymbol{\xi}(0)\|E_\alpha(-\gamma t^\alpha) + \varepsilon\left[\frac{k_\delta\lambda_M^{3/2}(P)}{\lambda_m^{1/2}(P)\lambda_m(Q)}\right], \tag{34}$$

and considering that $E_\alpha(-\gamma t^\alpha) \to 0$, one has that

$$\lim_{t\to\infty}\|\boldsymbol{\xi}(t)\| \le \varepsilon\left[\frac{k_\delta\lambda_M^{3/2}(P)}{\lambda_m^{1/2}(P)\lambda_m(Q)}\right], \tag{35}$$

establishing that the parameter ε modulates the radius of the domain of attraction for $\lim_{t\to\infty}\boldsymbol{\xi}$.

At this point, since one has that

$$\hat{d} = [k_p \quad -1]\boldsymbol{\xi} + d,$$

from Eqs. (22) to (23), it results

$$\begin{aligned}\lim_{t\to\infty}|\hat{d} - d| &\le \max(1, k_p)\lim_{t\to\infty}||\boldsymbol{\xi}|| \\ &\le \varepsilon\cdot\max(1,k_p)\left[\frac{k_\delta\lambda_M^{3/2}(P)}{\lambda_m^{1/2}(P)\lambda_m(Q)}\right]\end{aligned} \tag{36}$$

proving the assertion of the theorem. □

Remark 1. It is worth to comment that the peaking phenomenon, inherent to any high-gain scheme, is alleviated by means of the integral error definition of $\boldsymbol{\sigma}(t)$ since $\boldsymbol{\sigma}(0) = \mathbf{0}$. Besides, note that in the controller $\boldsymbol{u} = -K\boldsymbol{x} - \hat{\boldsymbol{d}}$, only the disturbance observer term $\hat{\boldsymbol{d}}$ depends explicitly on $1/\varepsilon$, and note also that $\hat{\boldsymbol{d}}(0) = 0$. Thus, the peaking phenomena reduced since $\boldsymbol{u}(0) = -K\boldsymbol{x}(0)$, without any dependence on $1/\varepsilon$. However, a very small ε could be catastrophic due to a limited sampling rate implementation and high-frequency noisy effects.

Theorem 2 implies that the observer $\hat{\boldsymbol{d}}$ estimates the disturbance $\boldsymbol{d}$ with a precision that can be modulated by means of ε. In addition, the precision of the convergence $\boldsymbol{x} \to \mathbf{0}$ can be directly adjusted by decreasing the value of ε, as it is stated in the following result.

Theorem 3. *Consider system (17) with the disturbance observer (22). Then, $\boldsymbol{x}$ converges inside an open vicinity of the origin, whose radius is proportional to ε.*

Proof. From Eq. (21), one has that $\mathscr{D}^\alpha\boldsymbol{\sigma} = -\hat{\boldsymbol{d}} + \boldsymbol{d} = \Delta\boldsymbol{d}$, and from Eqs. (22) to (23), one gets $\mathscr{D}^\alpha\boldsymbol{\sigma} = [-k_p \;\; 1]\boldsymbol{\xi}$, then $\Delta\boldsymbol{d} = [-k_p \;\; 1]\boldsymbol{\xi}$. Therefore Theorem 2 implies that $\Delta\boldsymbol{d}$ converges inside a vicinity of the origin, whose radius is proportional to ε. Thus, the precision of the approximation $\hat{\boldsymbol{d}} \to \boldsymbol{d}$ is directly proportional to ε.

The closed-loop system using the controller $\boldsymbol{u} = -K\boldsymbol{x} - \hat{\boldsymbol{d}}$ is

$$\dot{\boldsymbol{x}}(t) = \bar{A}\boldsymbol{x}(t) + B\Delta\boldsymbol{d}(t) \tag{37}$$

whose solution becomes

$$\boldsymbol{x}(t) = e^{\bar{A}t}\boldsymbol{x}(0) + \int_0^t e^{\bar{A}(t-\tau)}B\Delta\boldsymbol{d}(\tau)d\tau \tag{38}$$

whose behavior is directly related to the term $\Delta\boldsymbol{d}(\tau)$, which converges from the beginning to a compact vicinity of the origin. This means that is not possible for $\boldsymbol{x}(t)$ to escape in finite-time, and assuming the opposite leads to an absurd. In addition, since $\bar{A}$ is a Hurwitz matrix, there are symmetric and positive definite matrices $\bar{P}, \bar{Q} \in \mathbb{R}^{n\times n}$, such that

$$2\bar{Q} + \bar{P}\bar{A} + \bar{A}^T\bar{P} = 0, \tag{39}$$

and the candidate Lyapunov function $\bar{V} = \frac{1}{2}\boldsymbol{x}^T\bar{P}\boldsymbol{x}$ yields the following time derivative,

$$\dot{\bar{V}} = -\boldsymbol{x}^T\bar{Q}\boldsymbol{x} + \boldsymbol{x}^T\bar{P}B\Delta\boldsymbol{d} \tag{40}$$

Therefore, it is clear that

$$\|\boldsymbol{x}\| \geq \frac{\lambda_M(\bar{P})\lambda_M(B)}{\lambda_m(\bar{Q})}\|\Delta\boldsymbol{d}\| \Rightarrow \dot{\bar{V}} \leq 0. \tag{41}$$

From the expressions (35) and (36), for any $\eta > 0$, there is a finite time instant $t_\eta \geq 0$, such that $||\Delta\boldsymbol{d}|| < (1+\eta)\kappa\varepsilon$, with $\kappa = n\max(1, k_p)\left[\frac{k_\delta\lambda_M^{3/2}(P)}{\lambda_m^{1/2}(P)\lambda_m(Q)}\right]$. Hence, for any $t \geq t_\eta$, one has that

$$\|\boldsymbol{x}\| \geq \varepsilon\cdot(1+\eta)n\max(1,k_p)\left[\frac{\lambda_M(\bar{P})\lambda_M(B)}{\lambda_m(\bar{Q})}\cdot\frac{k_\delta\lambda_M^{3/2}(P)}{\lambda_m^{1/2}(P)\lambda_m(Q)}\right] \Rightarrow \dot{\bar{V}} \leq 0, \tag{42}$$

as a consequence, $\boldsymbol{x}$ converges inside an open vicinity of the origin, whose radius is proportional to ε. This produces that, by virtue of the action of the disturbance observer $\hat{\boldsymbol{d}}$, the state $\boldsymbol{x}$ evolves with a negligible effect of $\Delta\boldsymbol{d}$. Thus, the precision of the convergence of $\boldsymbol{x} \to \boldsymbol{0}$, via $\boldsymbol{u}_0 = -K\boldsymbol{x}$, can be directly adjusted by decreasing the value of ε. □

Remark 2. Note that the proposed controller $\boldsymbol{u} = \boldsymbol{u}_0 - \hat{\boldsymbol{d}}$ results as

$$\boldsymbol{u} = -K\boldsymbol{x} - \bar{k}_p\boldsymbol{\sigma} - \bar{k}_i\mathscr{I}^\alpha\boldsymbol{\sigma}, \tag{43}$$

which consists of a state feedback plus a fractional PI-like controller, with the proportional and integral feedback gains given by $\bar{k}_p = \dfrac{k_p}{\varepsilon}$ and $\bar{k}_i = \dfrac{k_i}{\varepsilon^2}$.

4. Simulations

A numerical study is programmed in Simulnk in Matlab, based on the Euler integrator running at 0.1 ms. The CRONE method was used with a transfer function of 10-order for a frequency range of $[10^{-3}, 10^3]$ rad/s [47].

Since the purpose of this simulation is to assess the reliability of the proposed disturbance observer, consider the simple but representative nonlinear pendulum system,

$$ml^2\ddot{q} = T - mgl\sin(q) + \varphi, \tag{44}$$

where $l = 0.5$ m is the length, $m = 0.5$ kg is the mass, $g = 9.81\,\mathrm{m/s^2}$ is the gravity, T is the control torque and φ is the unknown disturbance. For simulation purposes, consider $d = \varphi/(ml^2) = \sin(2t) + \sin(4t) + \text{noise}$, with $\text{noise} = \mathscr{I}^\beta rnd(\zeta)$ for $\beta = 0.8$, and rnd a random function within $[-100, 100]$ with sampling of 0.001 s, see Fig. 1.

The objective is to regulate at $q(t) = 0$ for any initial condition, even in the presence of the unknown disturbance d. After the change of variables $x_1 = q$, and $x_2 = \dot{q}$, system

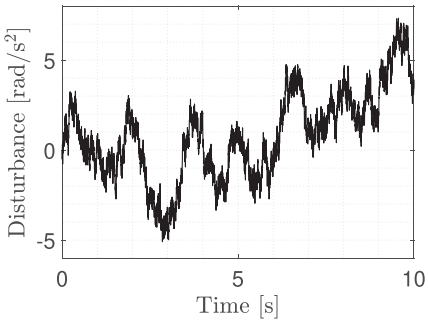
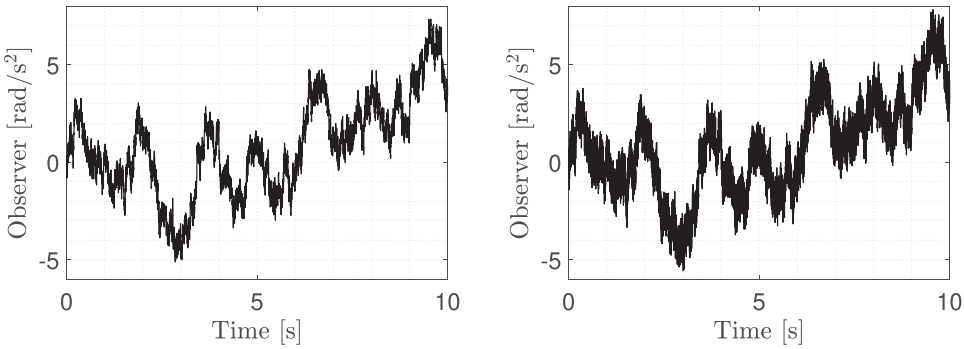


Fig. 1. Non-differentiable Hölder disturbance.



(a) High-gain observer (b) Sliding mode observer

Fig. 2. Disturbance observers.

(44) renders system (14) with $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The initial condition $q(0) = 2$ and $\dot{q}(0) = 0$ is considered.

It is evident that in the case of $d = 0$, the system is stabilised by means of the ideal controller $u_0 = -k_1 x_1 - k_2 x_2$, therefore, the integral error variable becomes

$$\sigma_0(t) = x_2(t) - x_2(0) + \int_0^t \Big(k_1 x_1(\tau) + k_2 x_2(\tau) \Big) d\tau \tag{45}$$

and $\sigma(t) = \mathscr{D}^{1-\alpha}\sigma_0(t)$ is the fractional deviation variable used to design the disturbance observer. The ideal control gains are set at $k_1 = 4$ and $k_2 = 4$, and those of the disturbance observer at $k_p = 2$ and $k_i = 1$.

Numerical results are shown for $\varepsilon = 0.002$ and $\alpha = 0.5$. Besides, the fractional sliding mode disturbance observer $\hat{d} = -k_d \mathscr{I}^{\alpha} \mathrm{sign}(\sigma_0)$, with $k_d = 25$ is considered for comparison purposes. It is worth noticing that the case $\alpha = 1$ is not considered since the studied disturbances are not assumed to be integer-order differentiable. The parameter ε of the high-gain observer was tuned by gradually reducing it to an allowable value in accordance with the

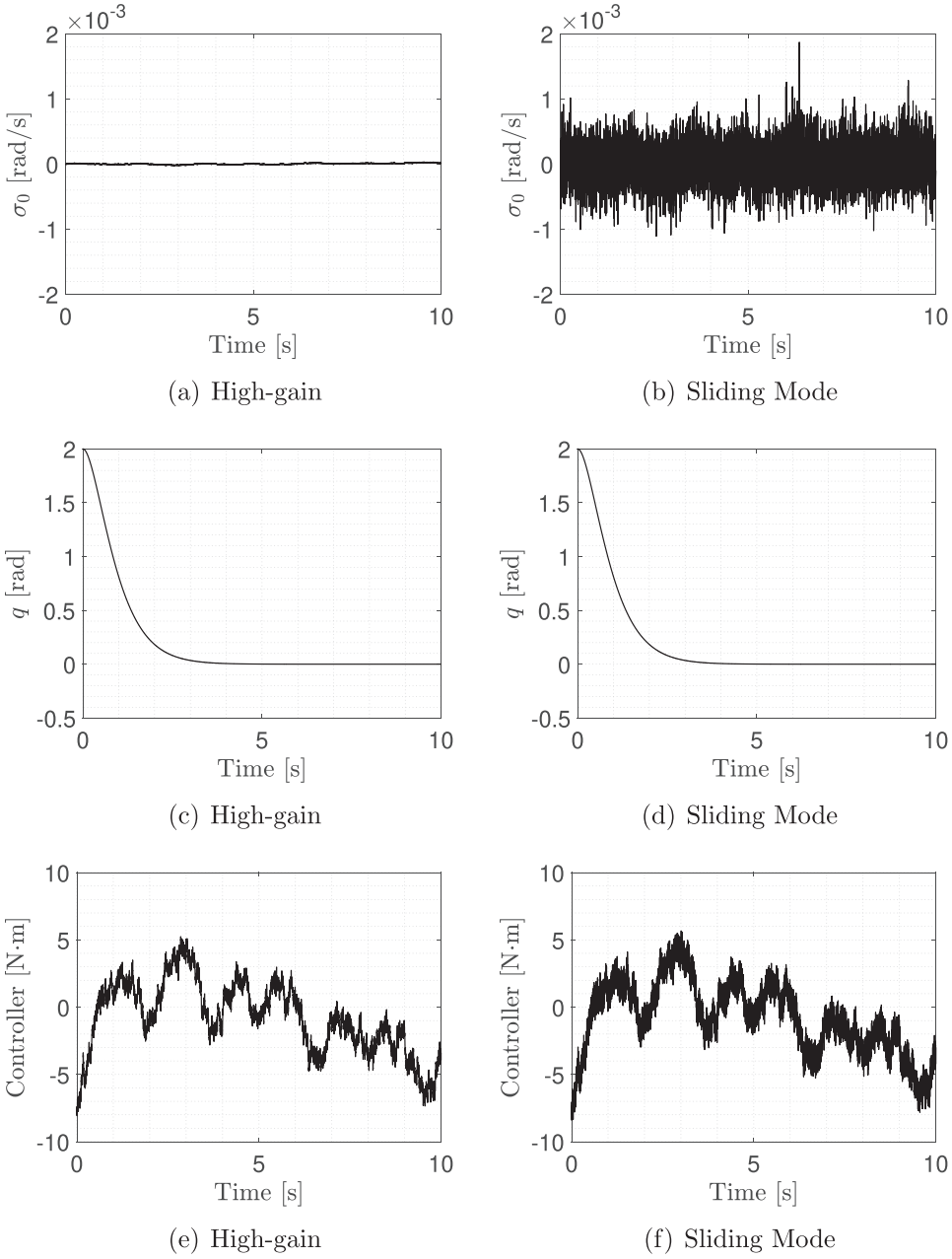


Fig. 3. Simulation results.

numerical stability of the simulator. On the other hand, gain k_d of the sliding mode observer was increased to achieve a good performance with small chattering.

Fig. 2 highlights the efficacy of both fractional-order disturbance observers, the high-gain linear and the sliding mode based schemes. A slightly improvement can be realized in the case of the high-gain approach.

In Fig. 3 one can appreciate that both, the high-gain and the sliding mode disturbance observer-based controllers yield acceptable performances since the variable q is regulated to the origin, with a high degree of precision. Nonetheless, the smoothness of the high-gain based controller produces a precise estimation, without any chattering effect, and consequently, a better closed-loop performance is obtained, as it can be figured out by observing the dynamics of σ_0.

5. Conclusion

This study constitutes an evidence that robust control tools in combination with fractional-order techniques furnish high-end structures to address a large class of physical phenomena in uncertain dynamical systems. It is shown that high-gain schemes outperform non-smooth techniques, such as sliding mode ones, through the capacity of modulating the convergence accuracy by means of a smooth controller. In addition, it is worth to comment that the peaking phenomenon, inherent to any high-gain scheme, is alleviated by means of the integral error, which, in the case of feedback systems, allows to account for the initial condition. Future studies consider the application of the proposed scheme in the control of uncertain dynamical systems, such as electromechanical servo-actuators design, robust robot control, stabilization of aerodynamical and underwater vehicles, and renewable energy systems, to name a few.

Declaration of Competing Interest

We authors declare that we have no interest conflict regarding the publication of this paper.

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